# Department of Management Studies

**Data Engineering End-To-End Cloud Project**

# "Azure Data Superstore Pipeline: End-to-End Data Engineering and Visualization"

**Program:** MBA-AI&DS

**Year & Semester:** III SEM

**Session: 2024-2025**


**Supervision by:**

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Assistant Professor, DOMS

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Enrollment No.: GE - 23144320

# Azure Data Superstore Pipeline: End-to-End Data Engineering and Visualization

**Introduction**

In today's data-driven world, the ability to process, analyze, and visualize large volumes of data efficiently is paramount. Organizations rely on seamless data pipelines to derive actionable insights that drive decision-making. This project is a practical implementation of a modern data engineering pipeline built using Microsoft Azure services. The goal is to process and analyze transactional sales data from the SampleSuperstore.csv dataset and present meaningful insights through interactive dashboards.

The project incorporates cutting-edge Azure technologies such as Azure Data Factory, Azure Data Lake Gen 2, Azure Synapse Analytics, and Azure Databricks, along with visualization tools like Power BI. By leveraging these tools, this project demonstrates the end-to-end process of data ingestion, transformation, storage, analysis, and visualization. It serves as a comprehensive example of how cloud-based solutions can streamline the data engineering lifecycle while ensuring scalability, reliability, and efficiency.

## Project Overview

This project demonstrates the implementation of a comprehensive data engineering pipeline leveraging Microsoft Azure services. The primary objective is to process and transform the SampleSuperstore.csv dataset, which contains transactional sales data, into actionable insights. This pipeline utilizes Azure Data Factory for data movement, Azure Data Lake Gen 2 for storage, Azure Synapse Analytics for querying and analysis, Azure Databricks for data transformation, and Power BI for visualization and reporting.

## Tools and Services Used

### Microsoft Azure Services

1. **Azure Data Factory (ADF):**
    - Orchestrates the movement and integration of data across various services.
    - Ensures seamless automation and scheduling of data workflows.

2. **Azure Data Lake Gen 2:**
    - Provides scalable and secure storage for structured and unstructured data.
    - Serves as the repository for raw and processed datasets.

3. **Azure Synapse Analytics:**
    - Facilitates querying and analysis of large datasets using SQL pools.
    - Supports advanced analytics and data warehousing capabilities.

4. **Azure Databricks:**
    - Enables distributed data processing and machine learning using Apache Spark.
    - Performs data cleansing, transformation, and enrichment.

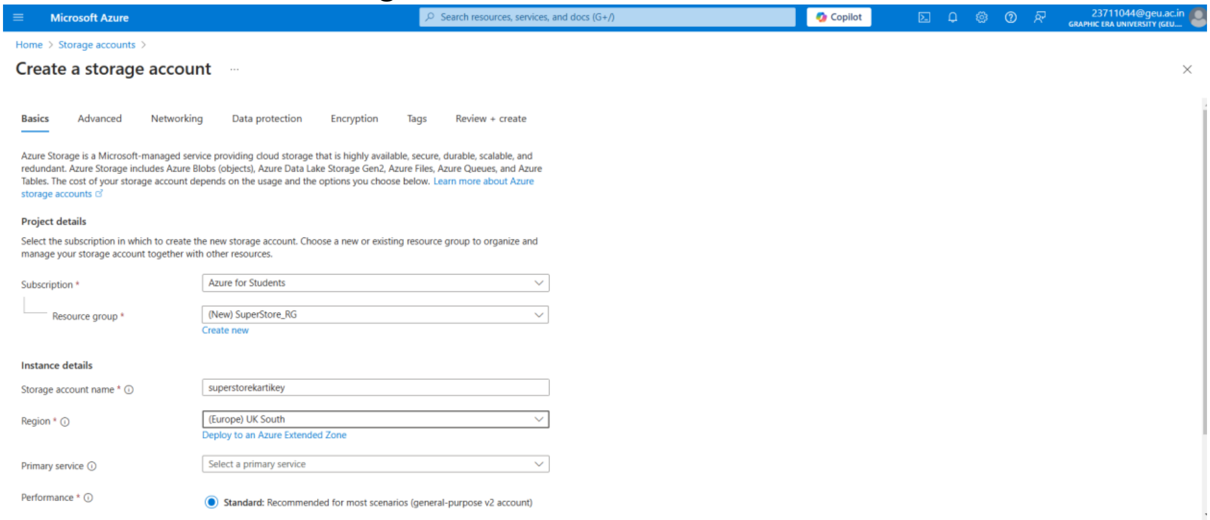
5. **Power BI:**
    - Visualizes data through interactive dashboards and reports.
    - Provides intuitive tools for business intelligence.

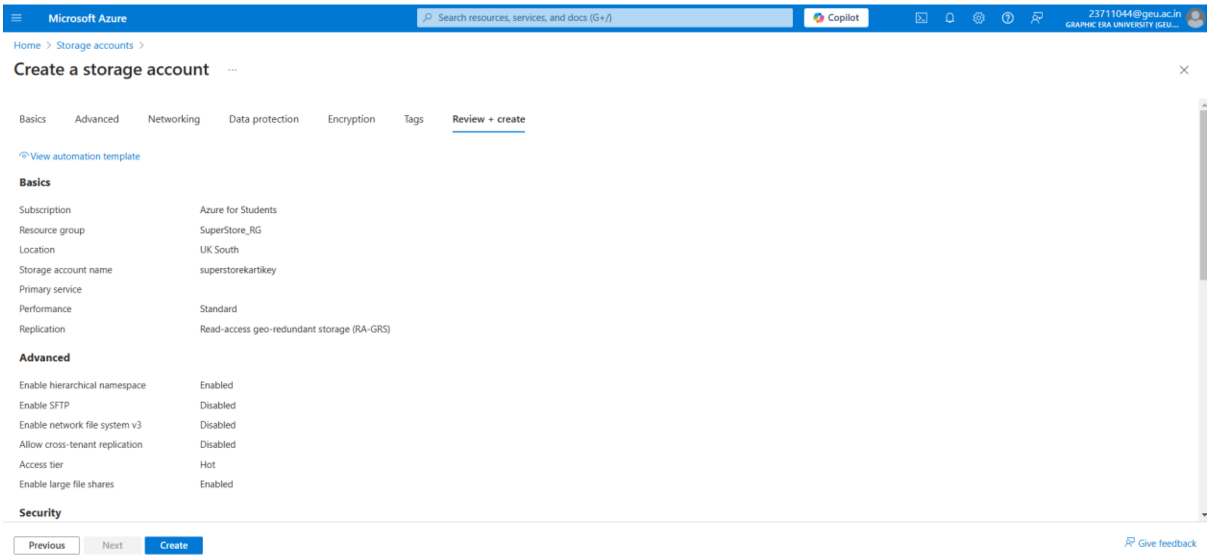
# Steps in the Data Pipeline:-

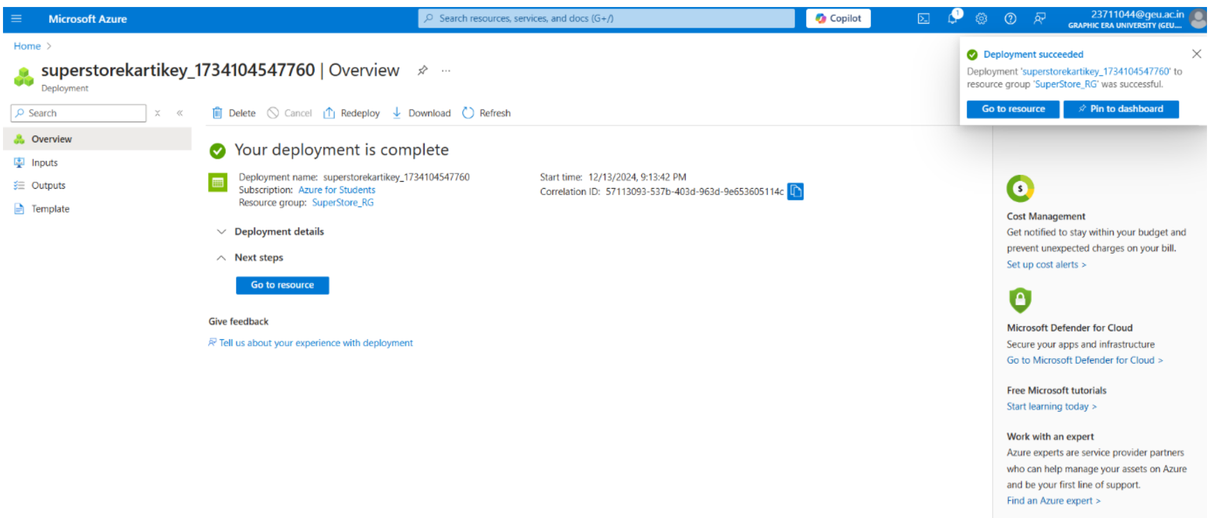
## Step 1: Data Collection and Storage

### 1. Setup Azure Data Lake Gen 2:

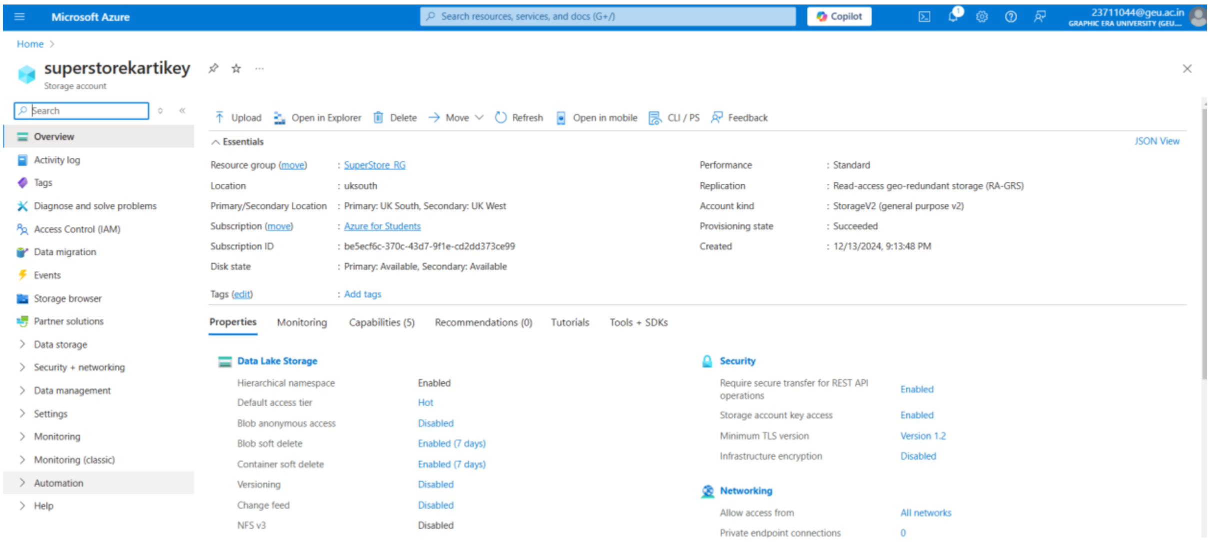
- Create a new storage account in Azure.

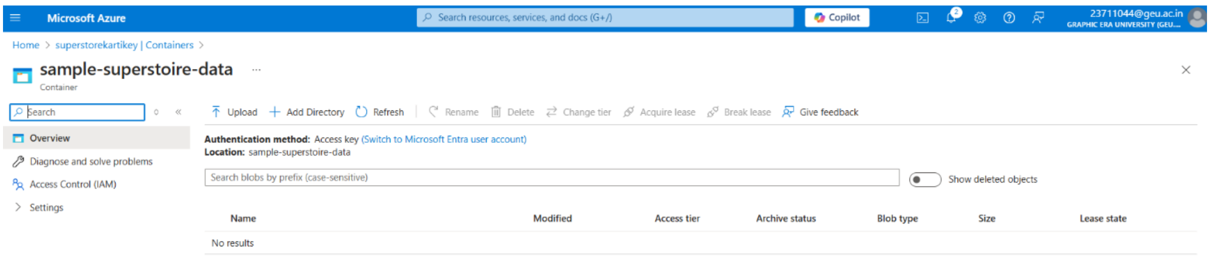


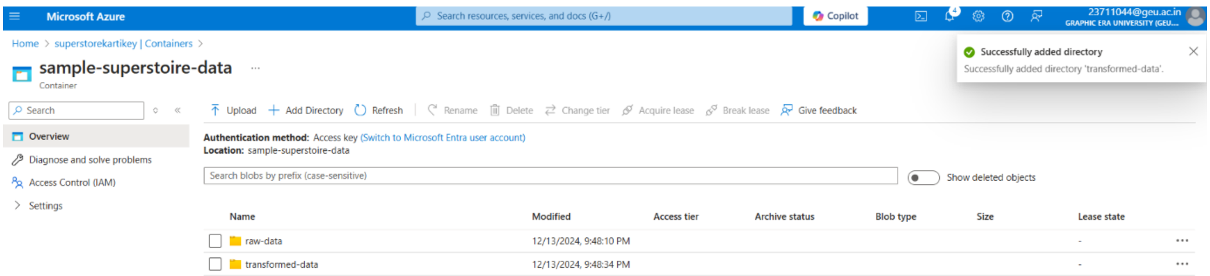



- Enable the hierarchical namespace to support Data Lake Gen 2 capabilities.
- Use Azure Storage Explorer to upload the SampleSuperstore.csv dataset into a designated container.



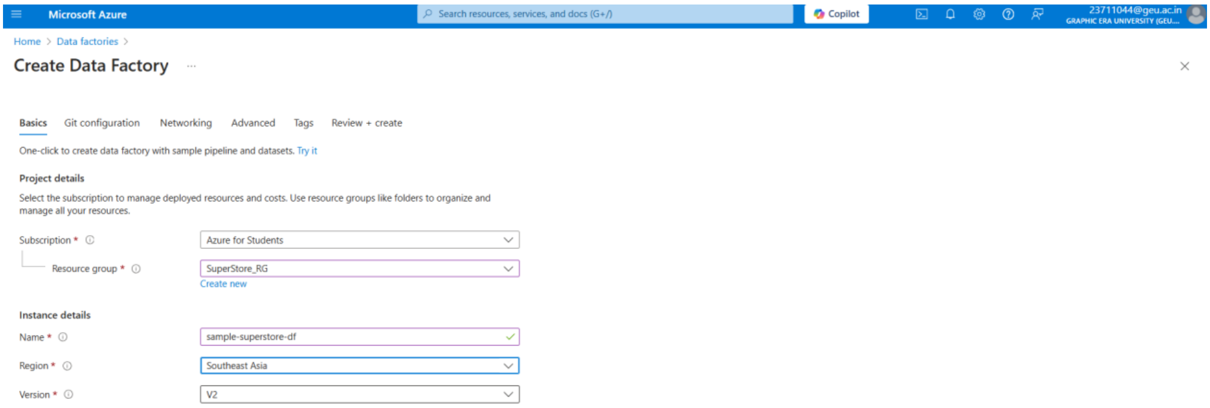



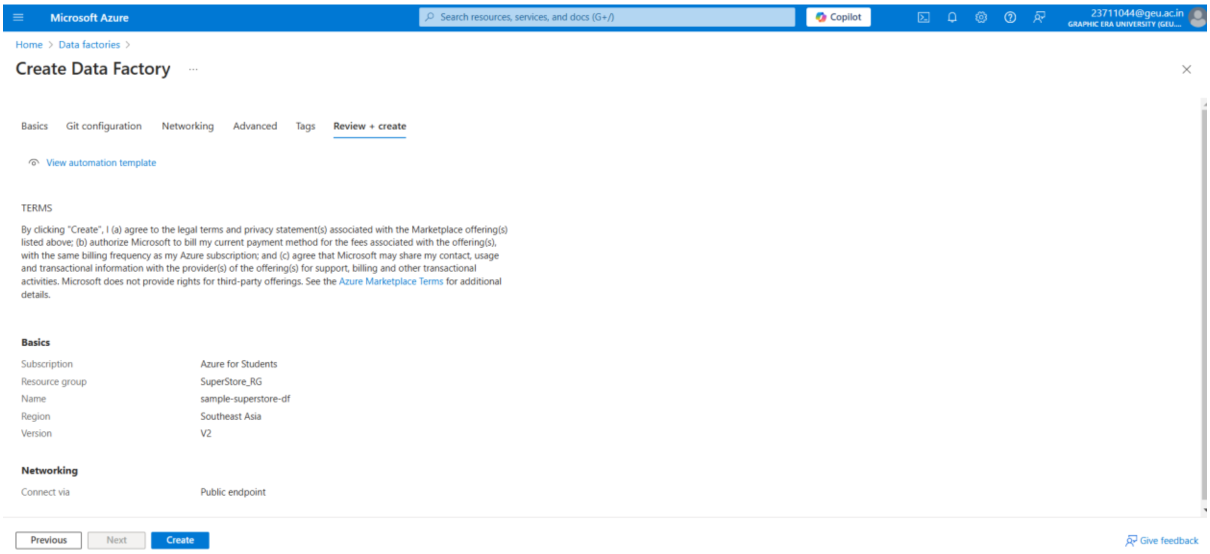

2. **Dataset Description:**
   - The SampleSuperstore.csv dataset contains columns such as Order Date, Region, Sales, Profit, and Category, providing transactional and geographic insights.

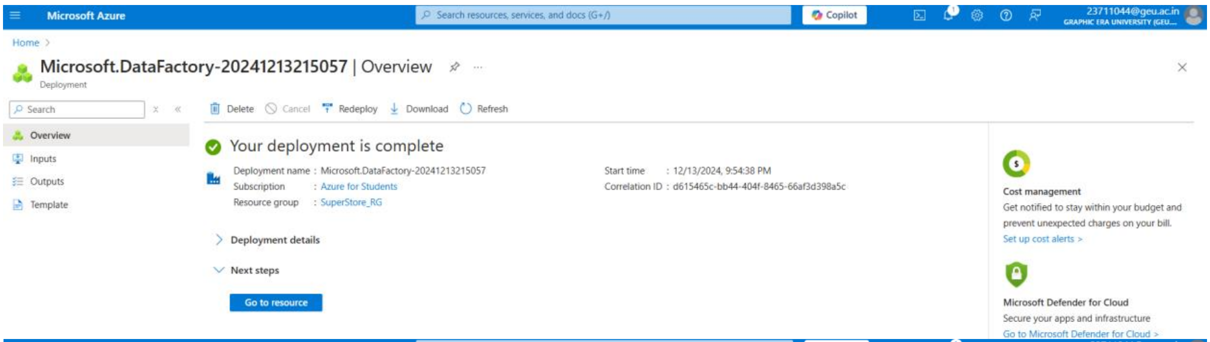
# Step 2: Data Movement Using Azure Data Factory

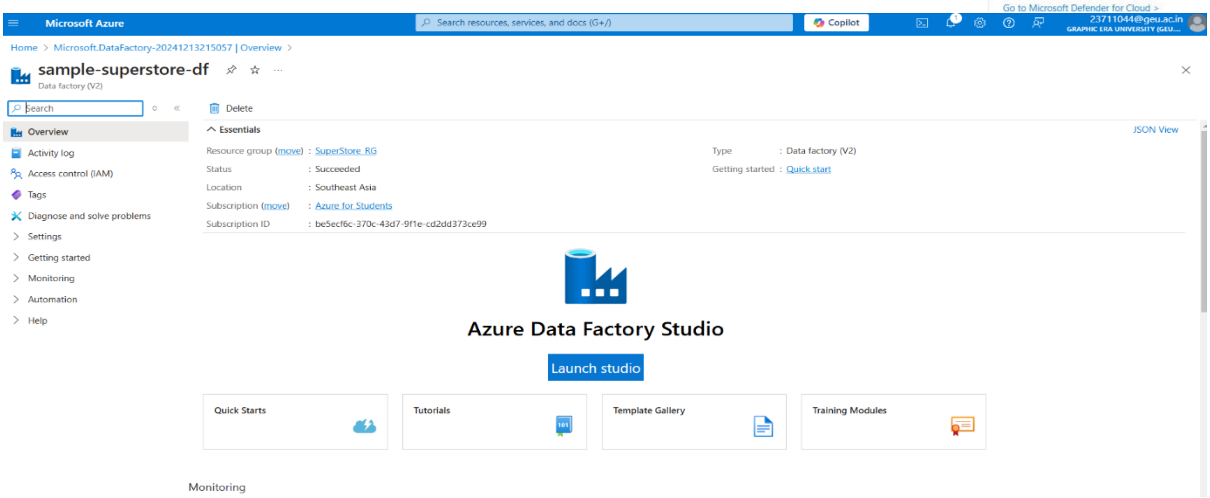
1. **Create Azure Data Factory Instance:**

   - Navigate to the Azure portal and set up a new Data Factory.

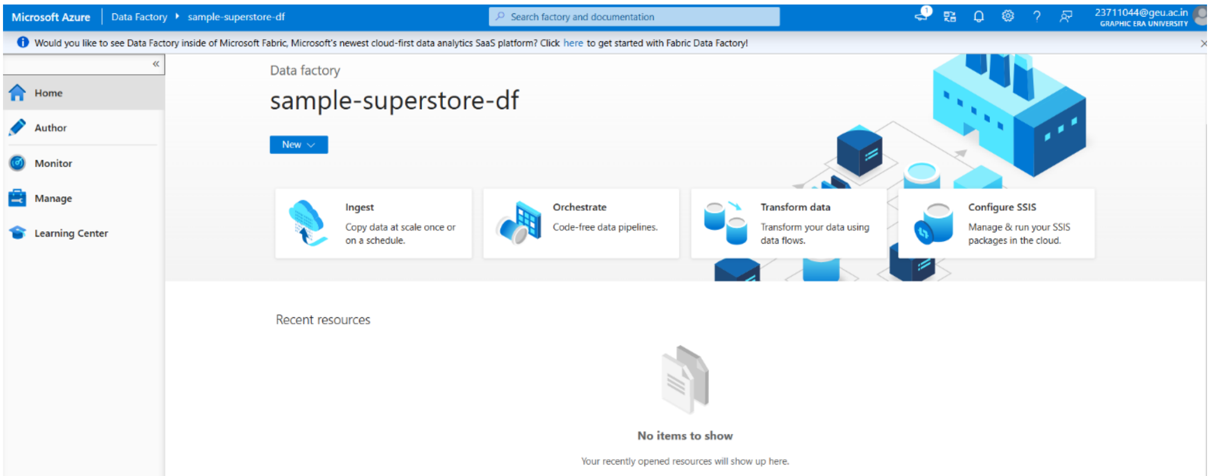


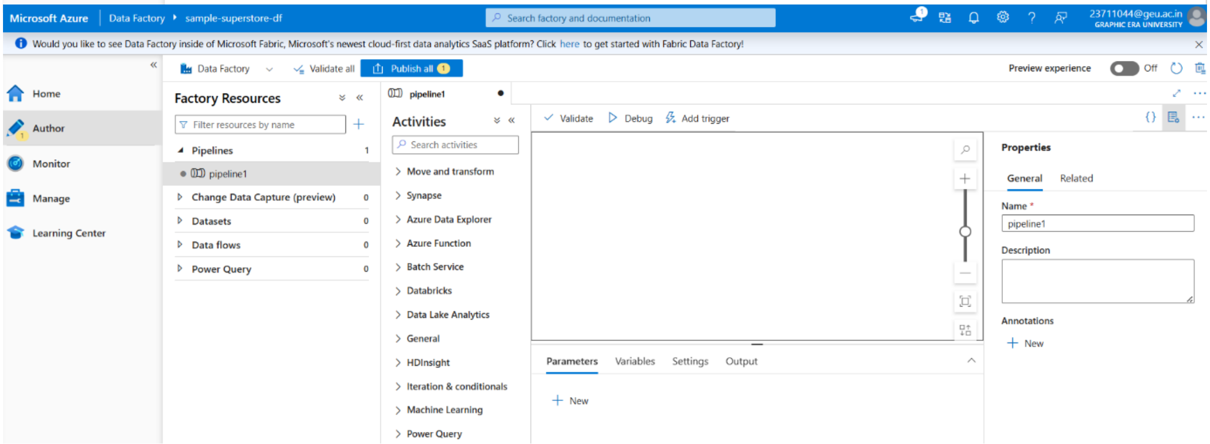



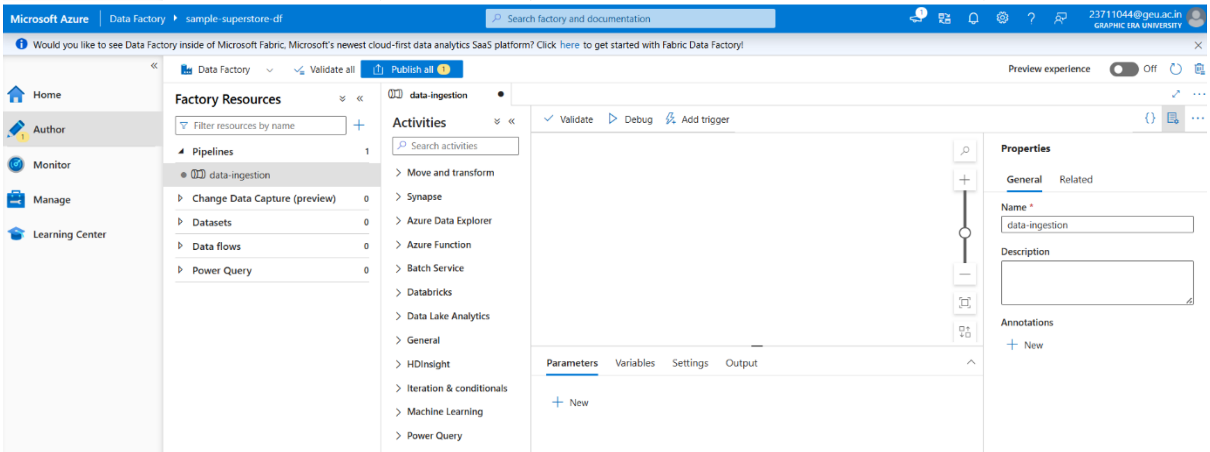



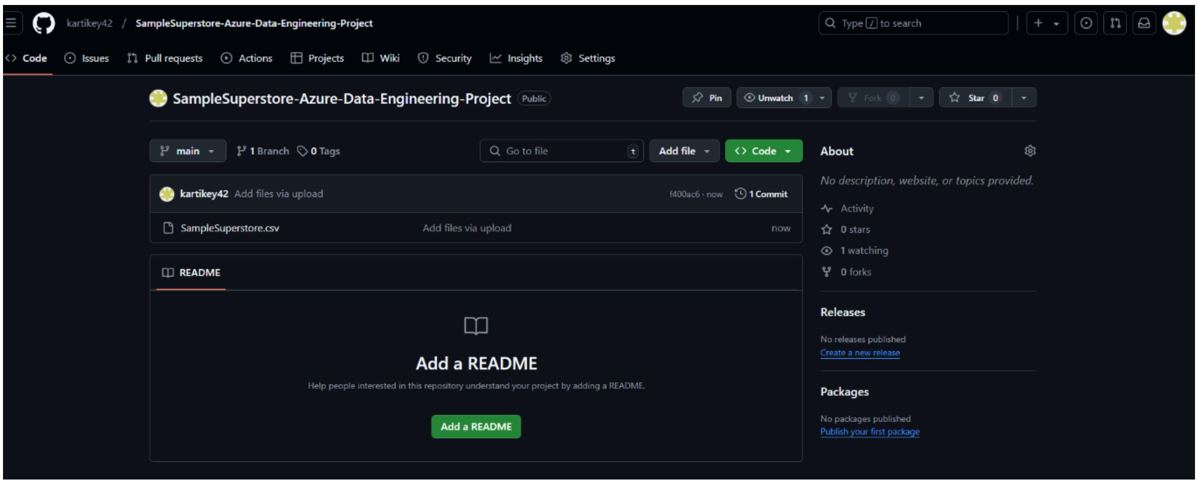



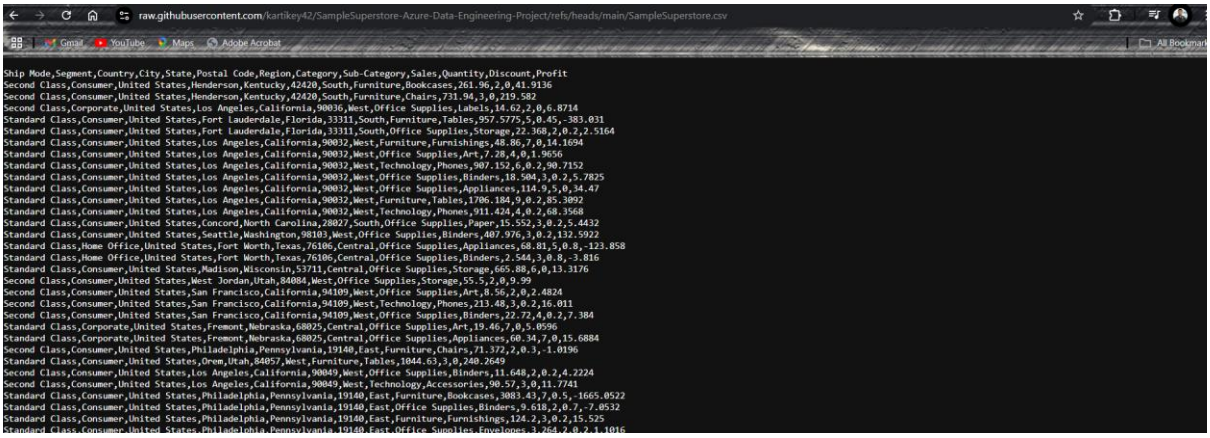
- Define linked services for both the source (Data Lake) and the target (Synapse Analytics).





raw.githubusercontent.com/kartikey42/SampleSuperstore-Azure-Data-Engineering-Project/refs/heads/main/SampleSuperstore.csv

```
Ship Mode,Segment,Country,City,State,Postal Code,Region,Category,Sub-Category,Sales,Quantity,Discount,Profit
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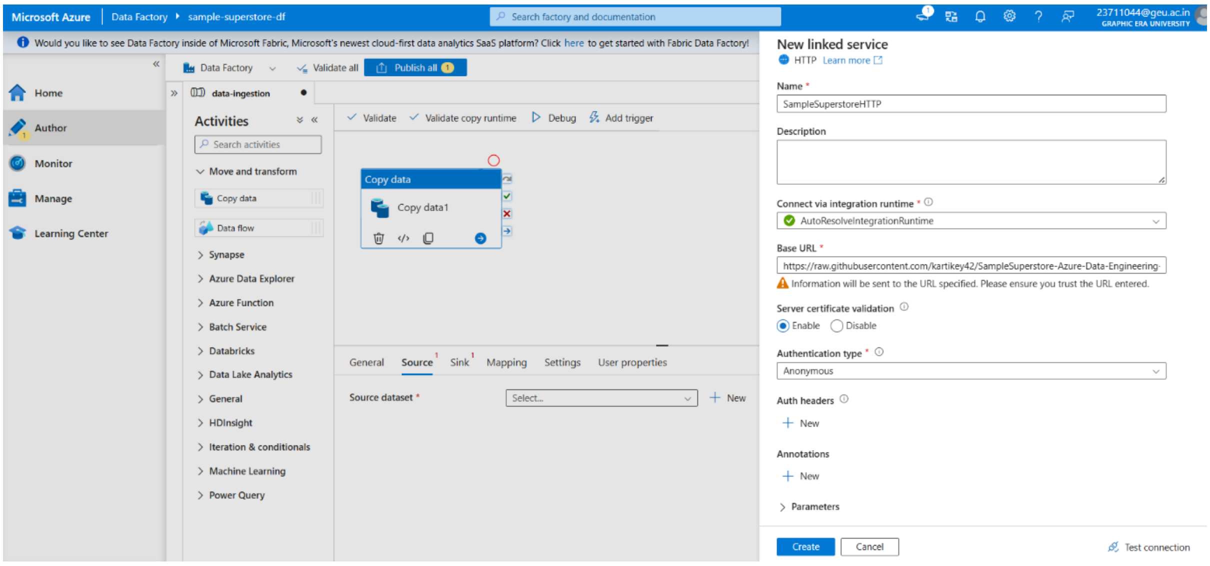


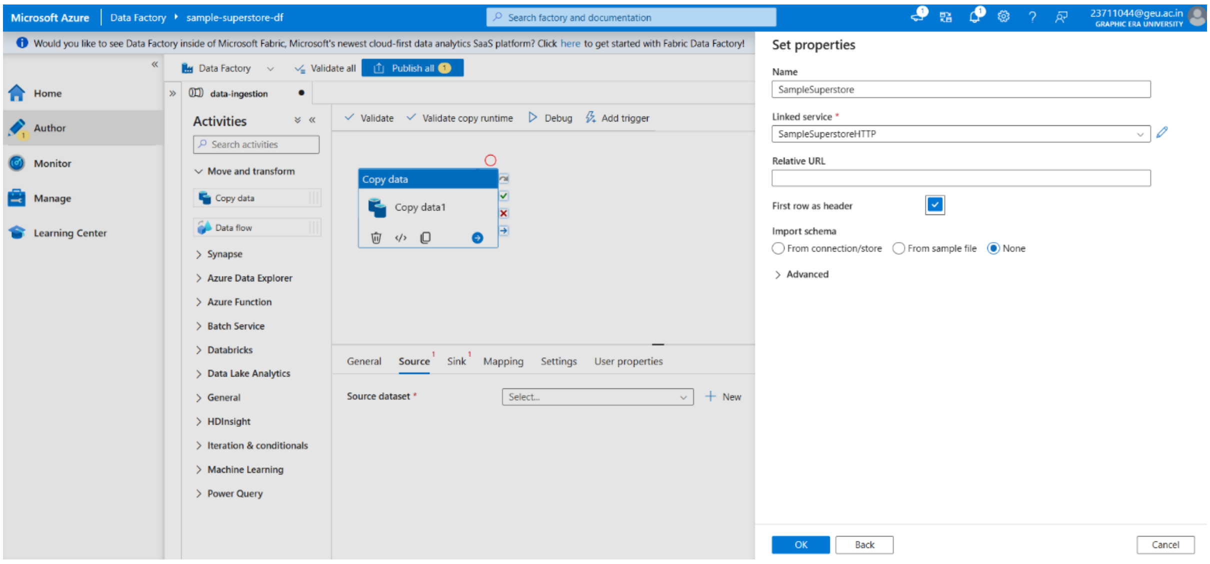


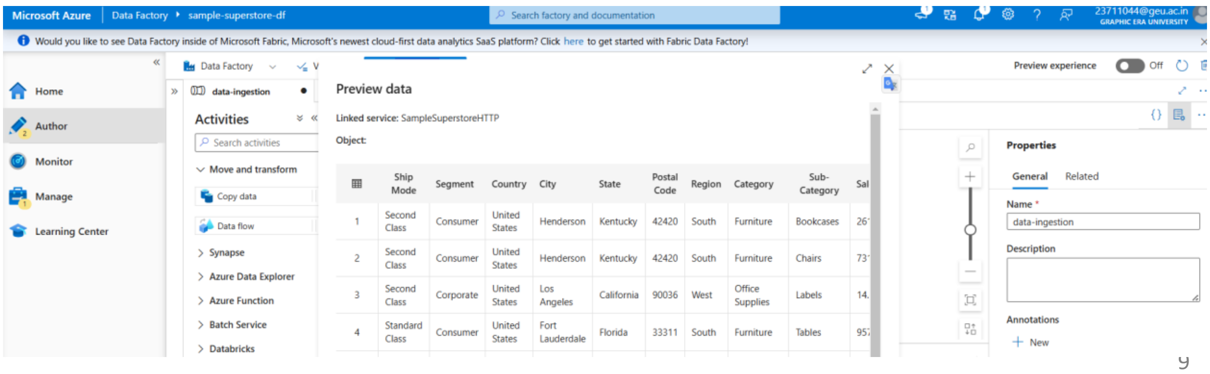

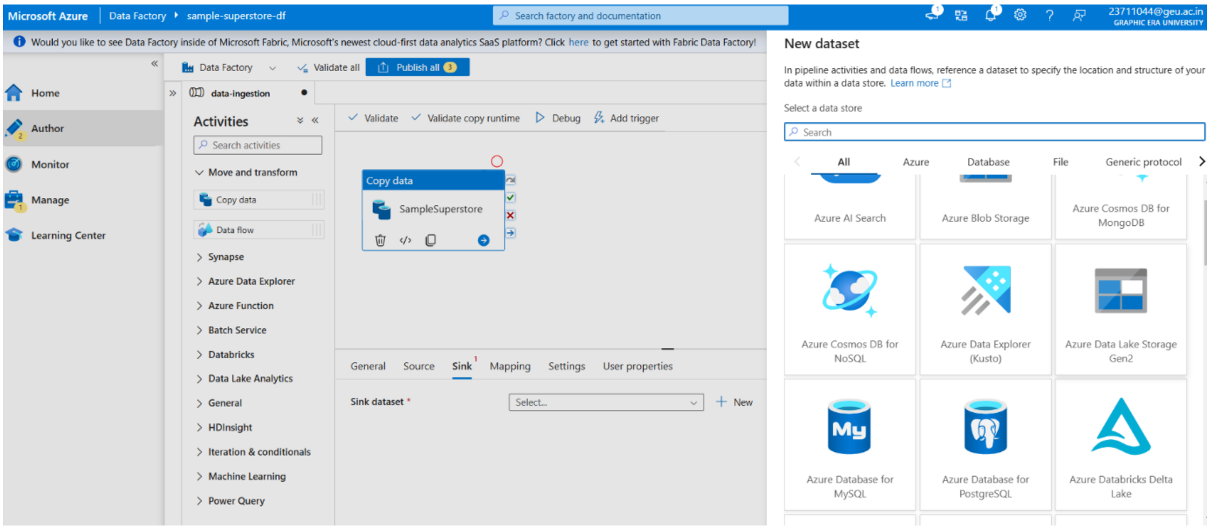
Microsoft Azure | Data Factory > sample-superstore-df
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Would you like to see Data Factory inside of Microsoft Fabric, Microsoft's newest cloud-first data analytics SaaS platform? Click here to get started with Fabric Data Factory!
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Validate
Validate copy runtime
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Copy data
SampleSuperstore
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Source
Sink
Mapping
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User properties
Sink dataset *
Select...
New
New dataset
In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more
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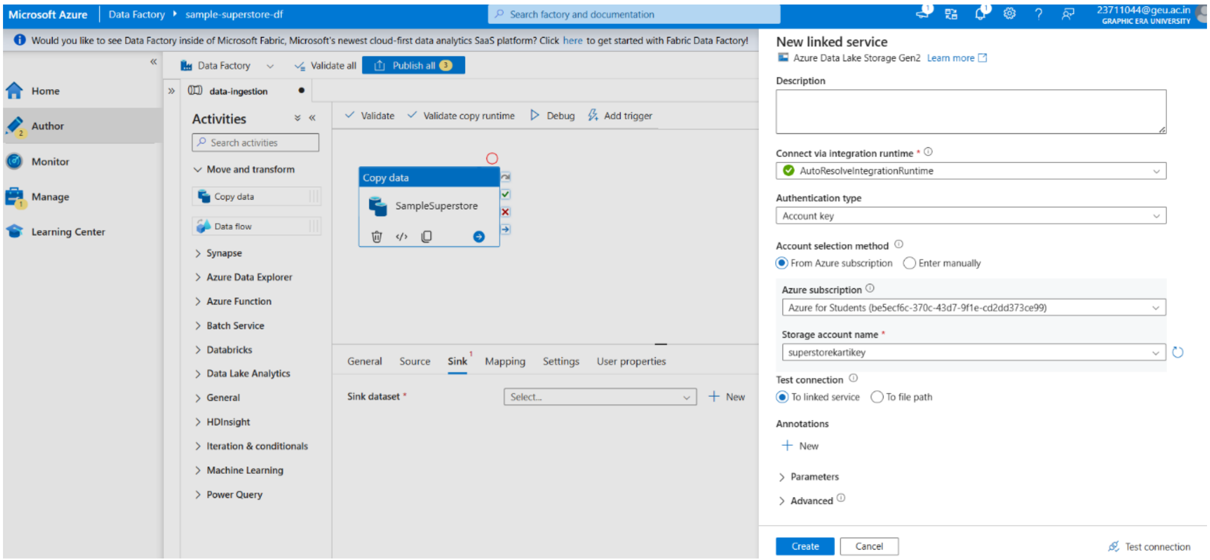
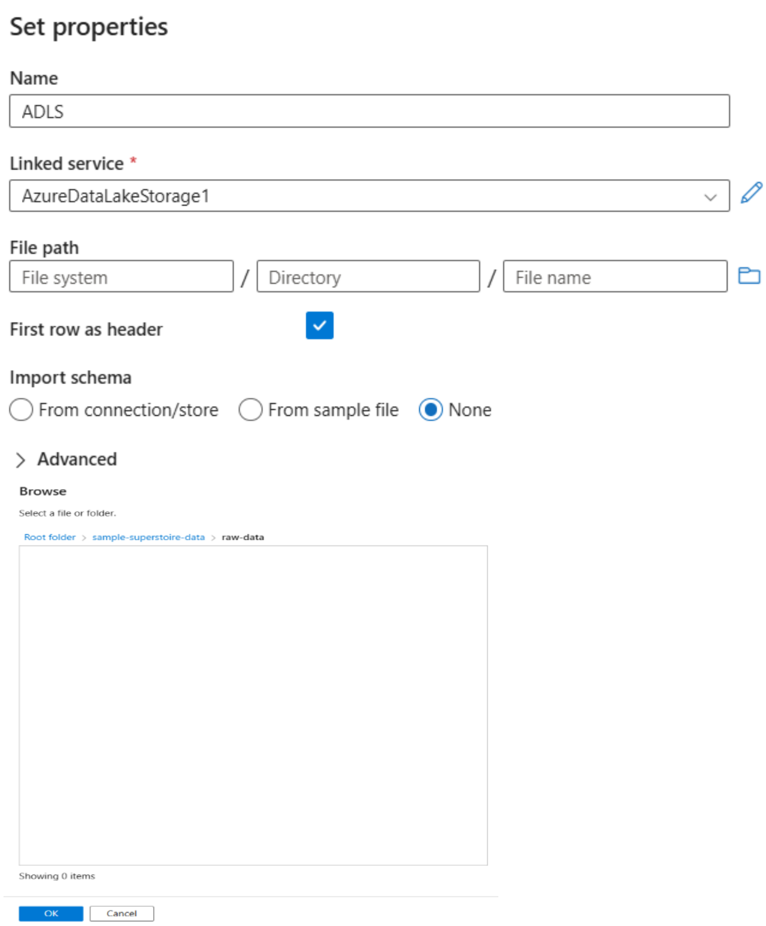
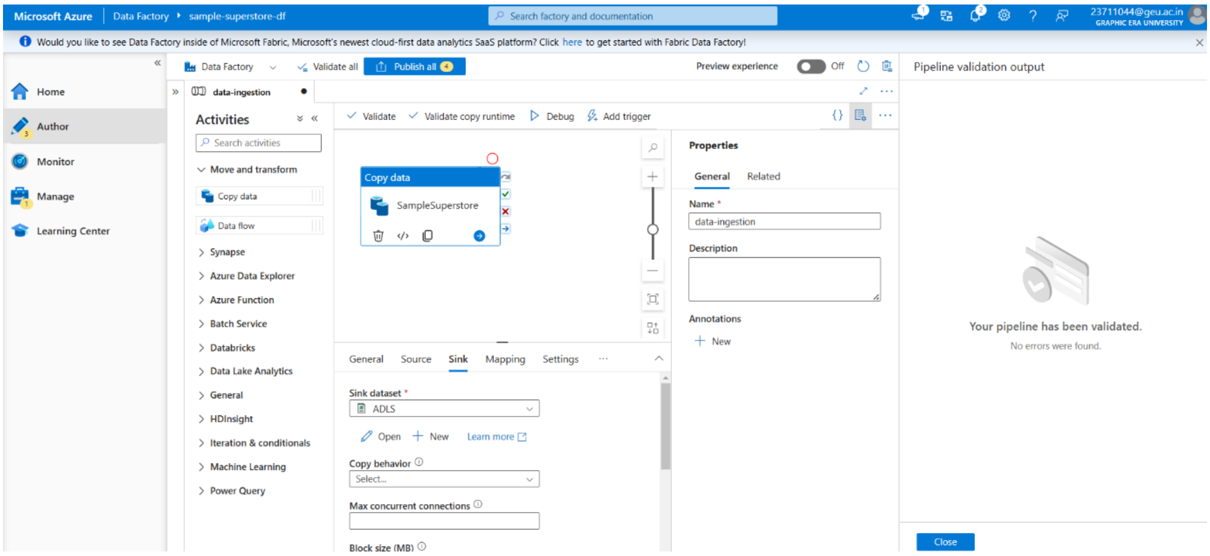
Search
All
Azure
Database
File
Generic protocol
Azure AI Search
Azure Blob Storage
Azure Cosmos DB for MongoDB
Azure Cosmos DB for NoSQL
Azure Data Explorer (Kusto)
Azure Data Lake Storage Gen2
Azure Database for MySQL
Azure Database for PostgreSQL
Azure Databricks Delta Lake

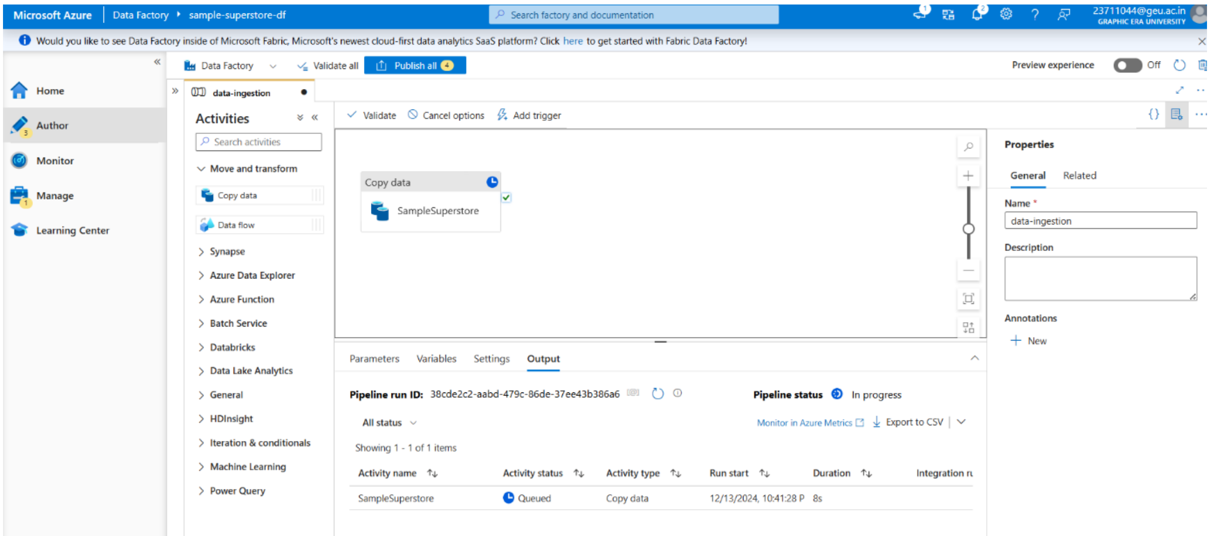

Microsoft Azure | Data Factory > sample-superstore-df
New linked service
Azure Data Lake Storage Gen2 Learn more
Description
Connect via integration runtime *
AutoResolveIntegrationRuntime
Authentication type
Account key
Account selection method
From Azure subscription
Enter manually
Azure subscription
Azure for Students (be5ecf6c-370c-43d7-9f1e-cd2dd373ce99)
Storage account name *
superstorekartikey
Test connection
To linked service
To file path
Annotations
New
Parameters
Advanced
Create
Cancel
Test connection

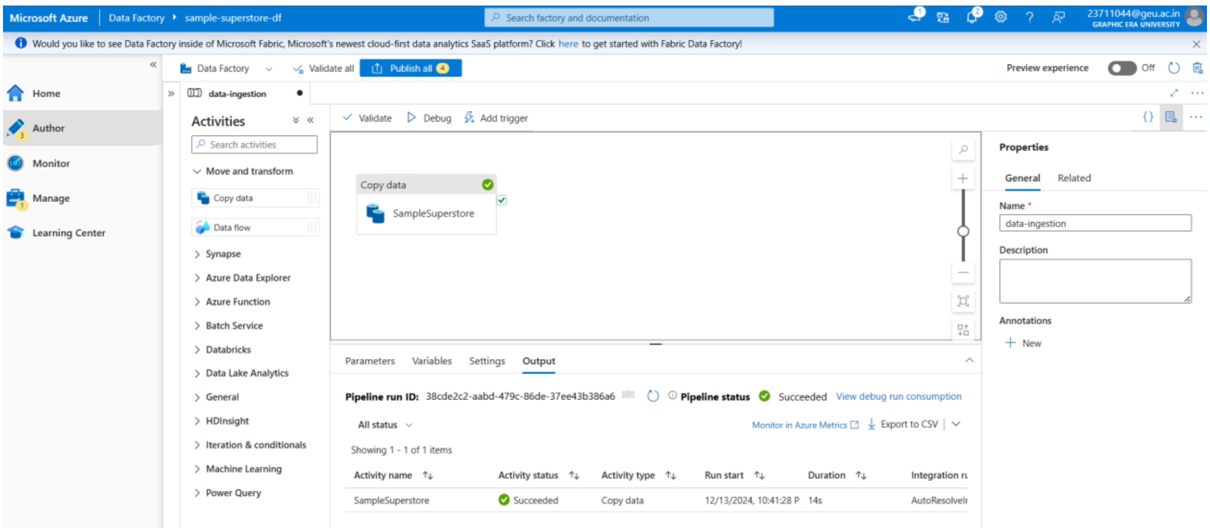

Set properties
Name
ADLS
Linked service *
AzureDataLakeStorage1
File path
File system
Directory
File name
First row as header
Import schema
From connection/store
From sample file
None
Advanced
Browse
Select a file or folder.
Root folder > sample-superstoire-data > raw-data
Showing 0 items
OK
Cancel



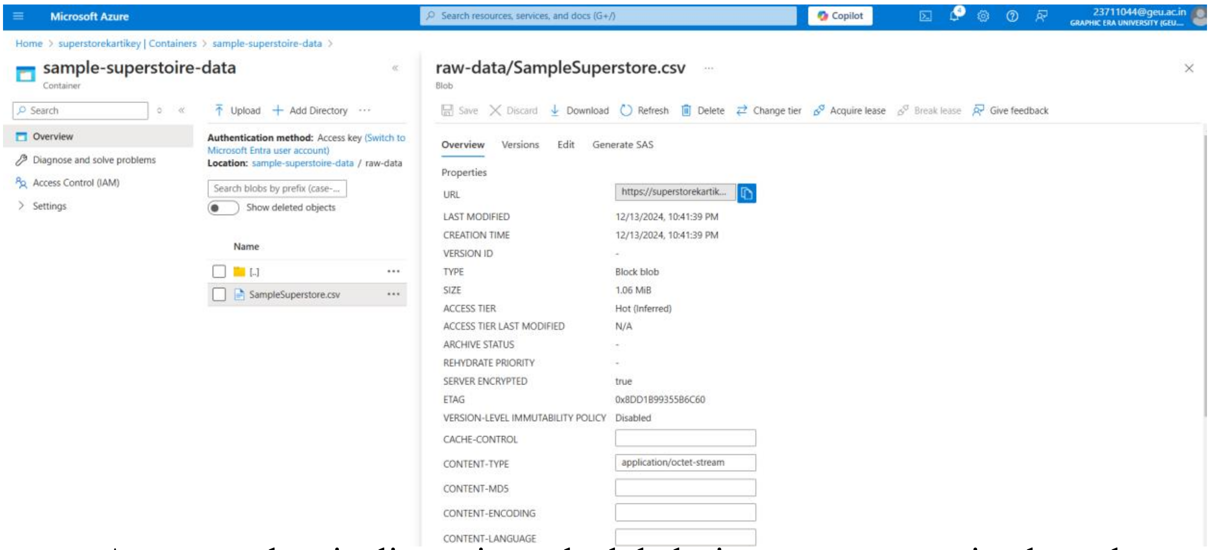

2. **Pipeline Design:**

- Use the **Copy Data** activity to transfer data from the Data Lake to Synapse Analytics.

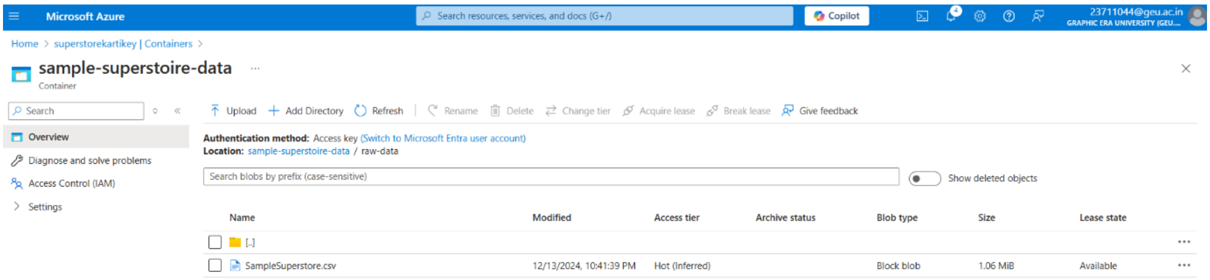


- Configure additional transformations such as column mapping or filtering

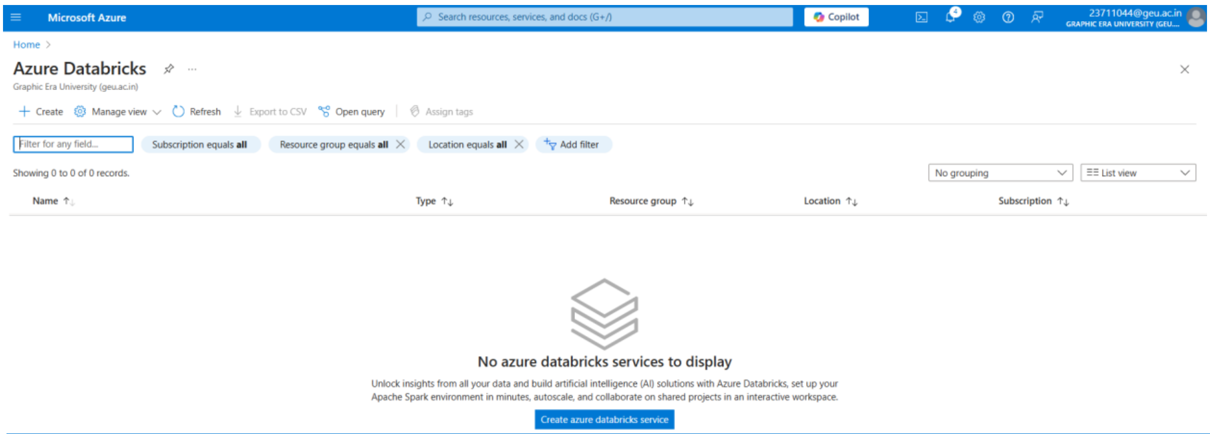


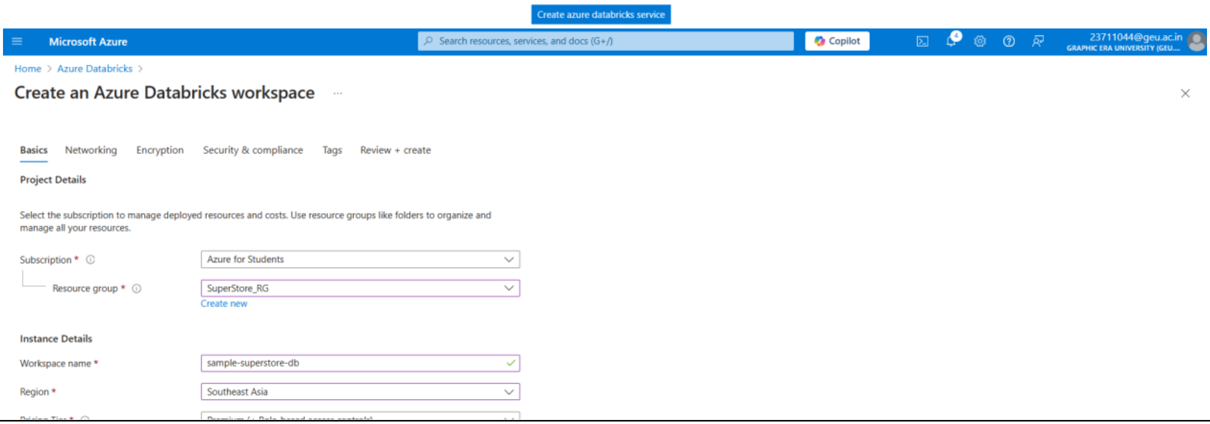

- Automate the pipeline using scheduled triggers to ensure timely updates.

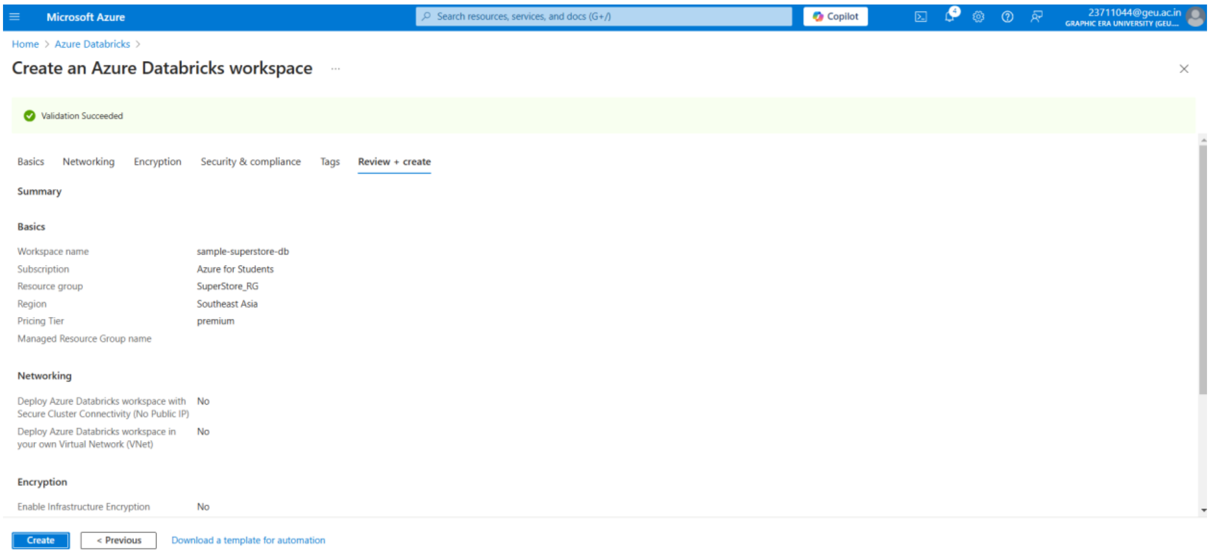


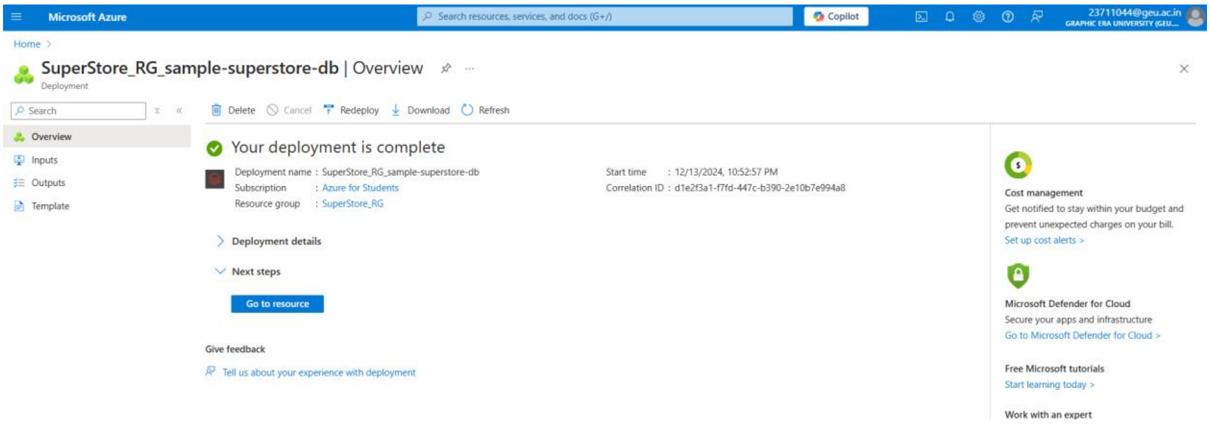

## Step 3: Data Transformation Using Azure Databricks

1. **Setup Databricks Workspace:**
   - Create a Databricks workspace and configure a Spark cluster.

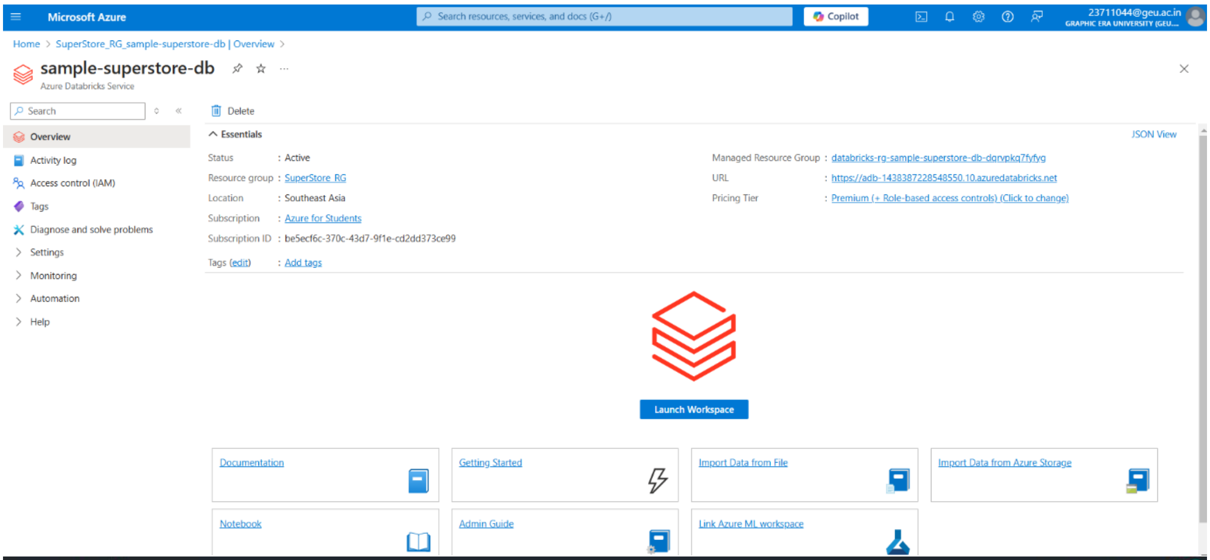


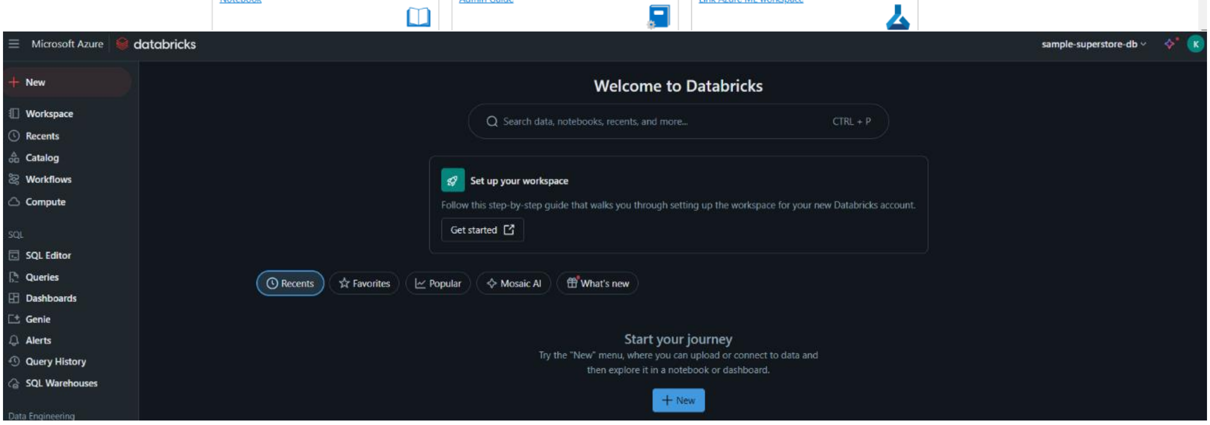



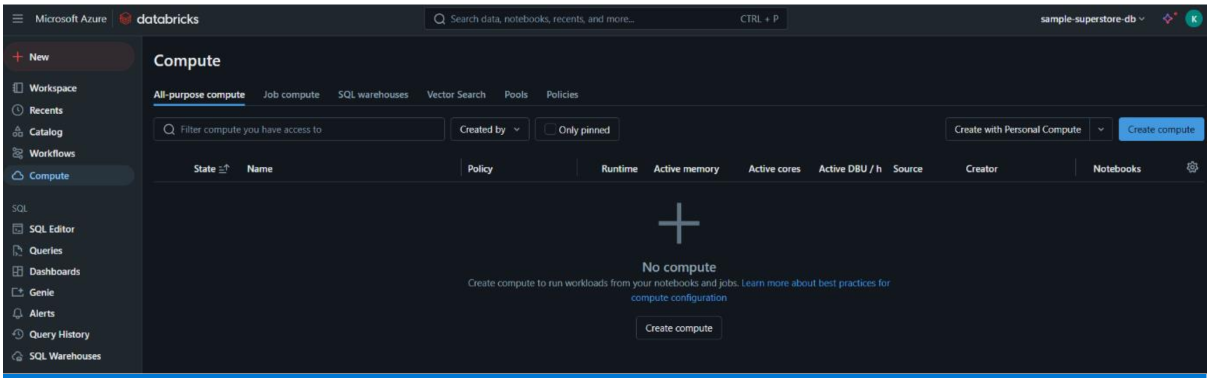



- Attach the cluster to the workspace for distributed processing.




Microsoft Azure
databricks
Search data, notebooks, recents, and more...
CTRL + P
sample-superstore-db
New
Workspace
Recents
Catalog
Workflows
Compute
SQL
SQL Editor
Queries
Dashboards
Genie
Alerts
Query History
SQL Warehouses
Compute
All-purpose compute
Job compute
SQL warehouses
Vector Search
Pools
Policies
Filter compute you have access to
Created by
Only pinned
Create with Personal Compute
Create compute
State
Name
Policy
Runtime
Active memory
Active cores
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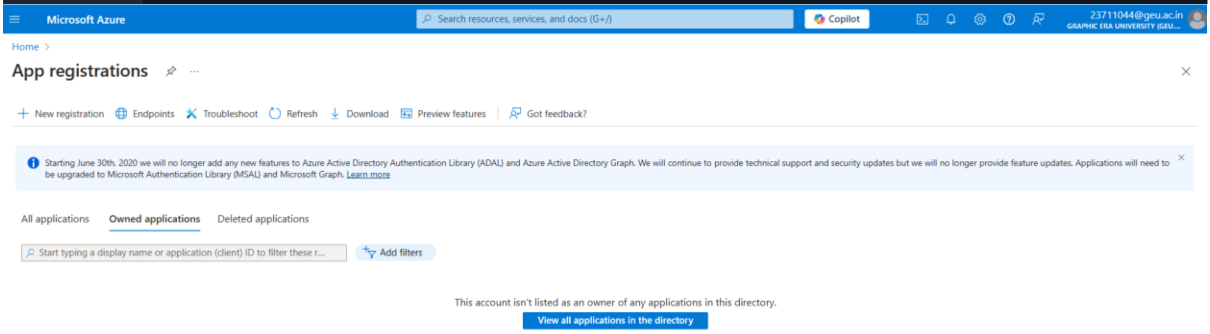
Notebooks
No compute
Create compute to run workloads from your notebooks and jobs. Learn more about best practices for compute configuration
Create compute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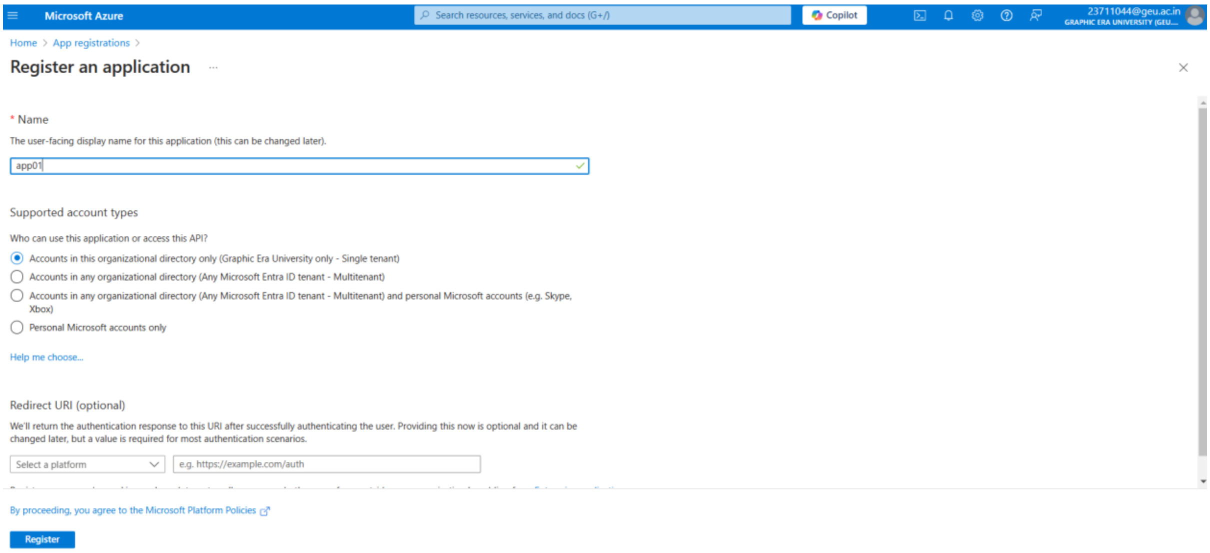
Start typing a display name or application (client) ID to filter these r...
Add filters
This account isn't listed as an owner of any applications in this directory.
View all applications in the directory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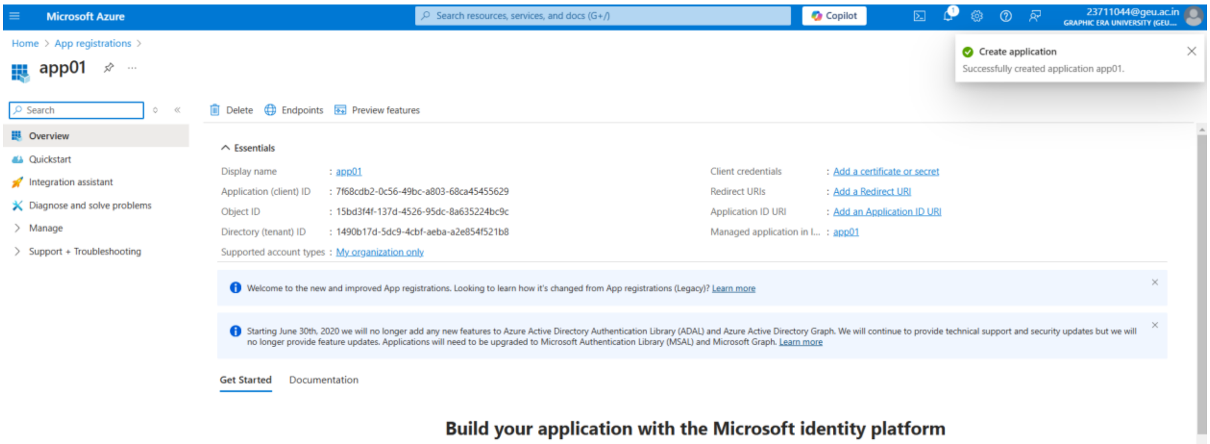
Home
App registrations
Register an application
* Name
The user-facing display name for this application (this can be changed later).
app01
Supported account types
Who can use this application or access this API?
Accounts in this organizational directory only (Graphic Era University only - Single tenant)
Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant)
Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant) and personal Microsoft accounts (e.g. Skype, Xbox)
Personal Microsoft accounts only
Help me choose...
Redirect URI (optional)
We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.
Select a platform
e.g. https://example.com/auth
By proceeding, you agree to the Microsoft Platform Policies
Register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app01
Create application
Successfully created application app01.
Search
Overview
Quickstart
Integration assistant
Diagnose and solve problems
Manage
Support + Troubleshooting
Delete
Endpoints
Preview features
Essentials
Display name : app01
Application (client) ID : 7f68cdb2-0c56-49bc-a803-68ca45455629
Object ID : 15bd3f4f-137d-4526-95dc-8a635224bc9c
Directory (tenant) ID : 1490b17d-5dc9-4cbf-aeba-a2e854f521b8
Supported account types : My organization only
Client credentials : Add a certificate or secret
Redirect URIs : Add a Redirect URI
Application ID URI : Add an Application ID URI
Managed application in l... : app01
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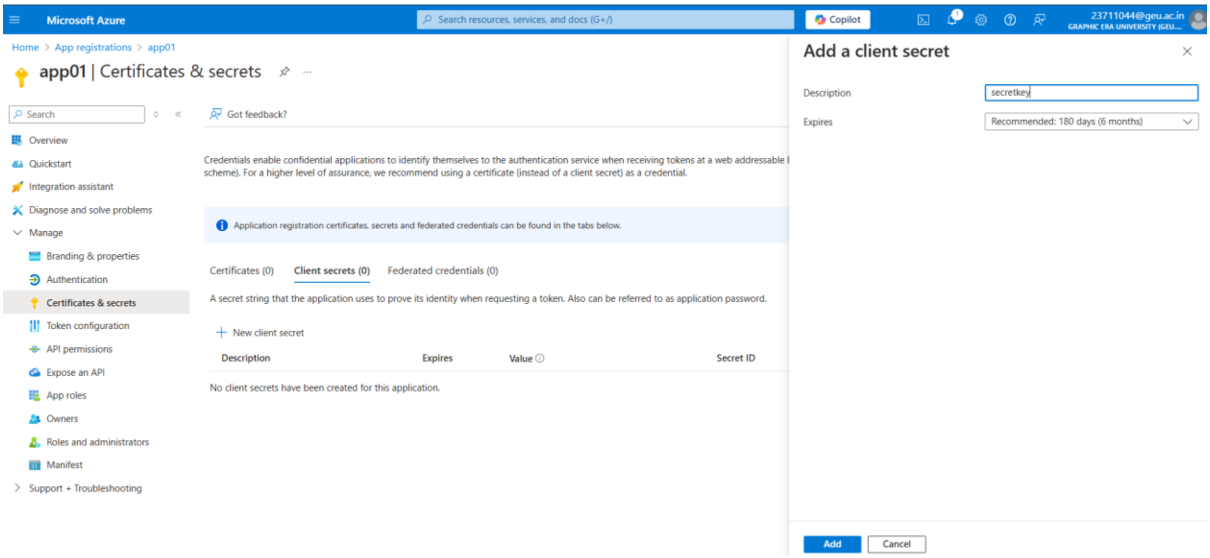
Get Started
Documentation
Build your application with the Microsoft identity platform

2. **Data Processing:**

- Mount the Azure Data Lake Gen 2 storage to the Databricks environment for seamless data access.

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Directory (tenant) ID : 1490b17d-5dc9-4cbf-aeba-a2e854f521b8

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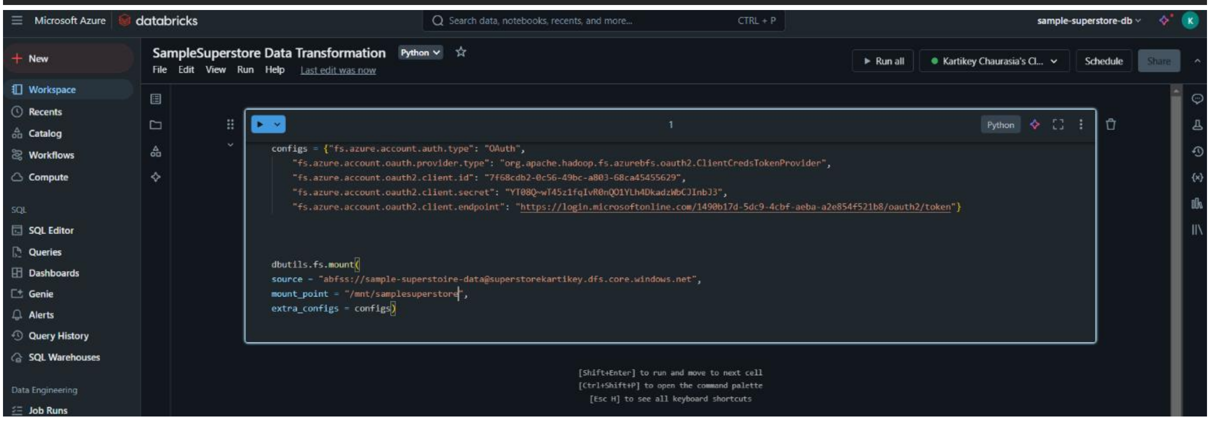
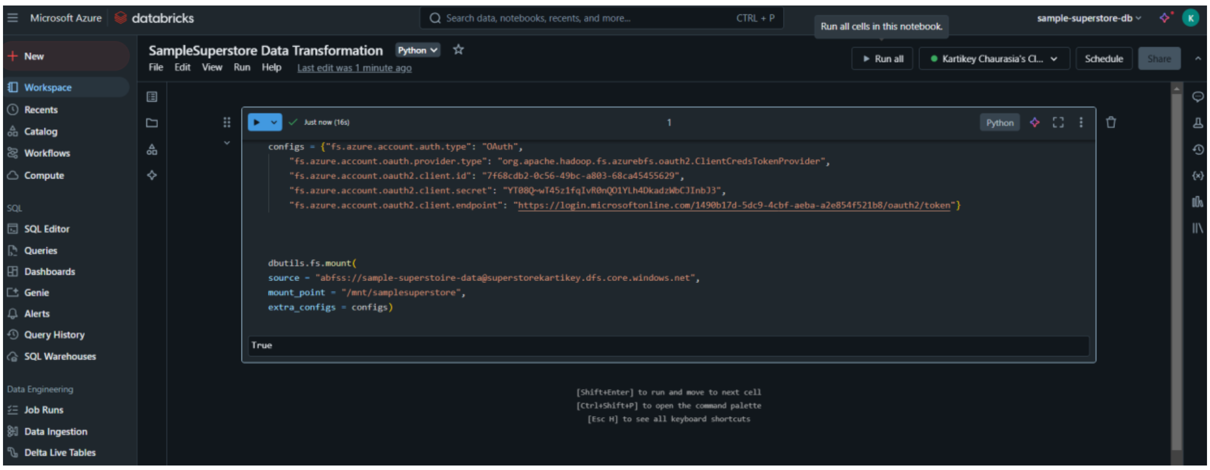
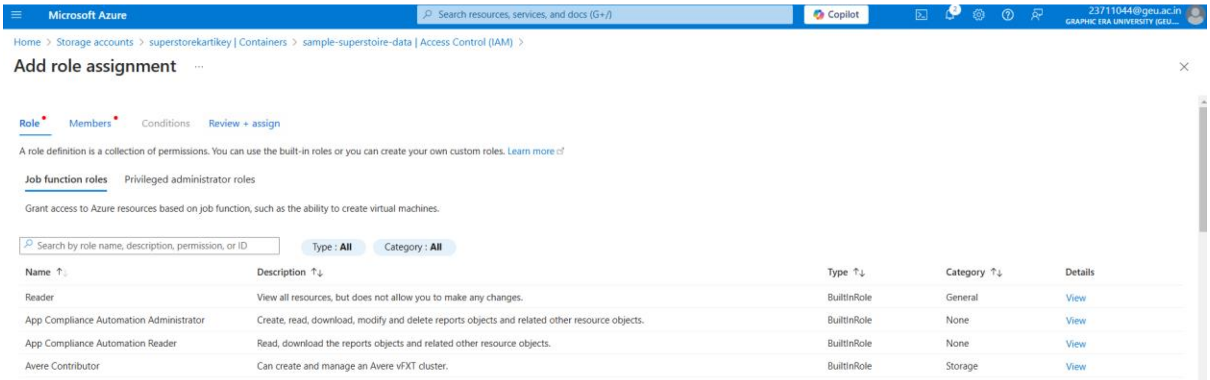
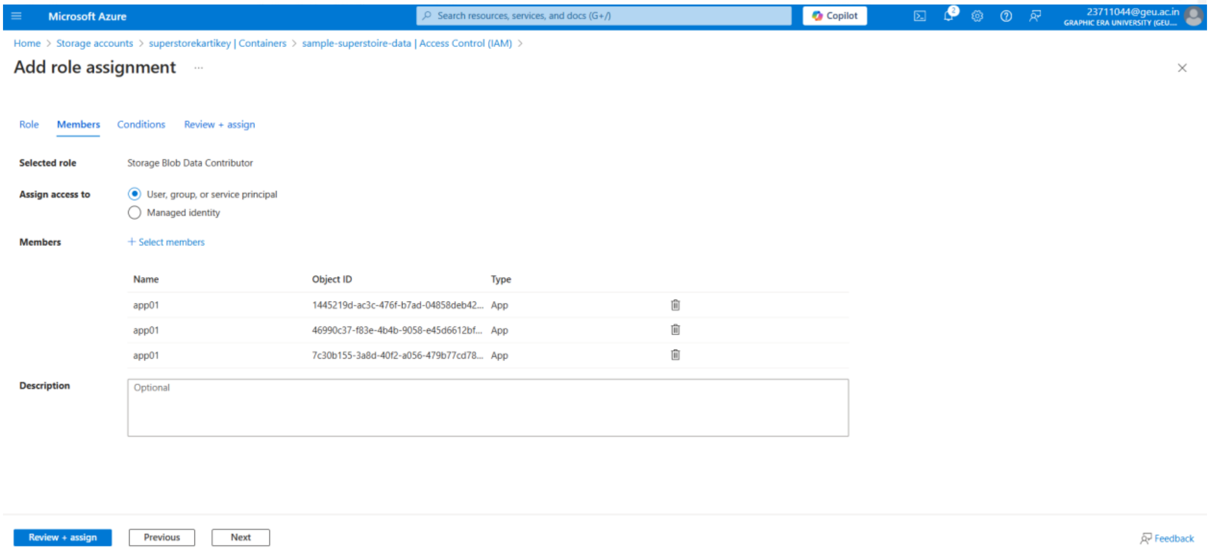
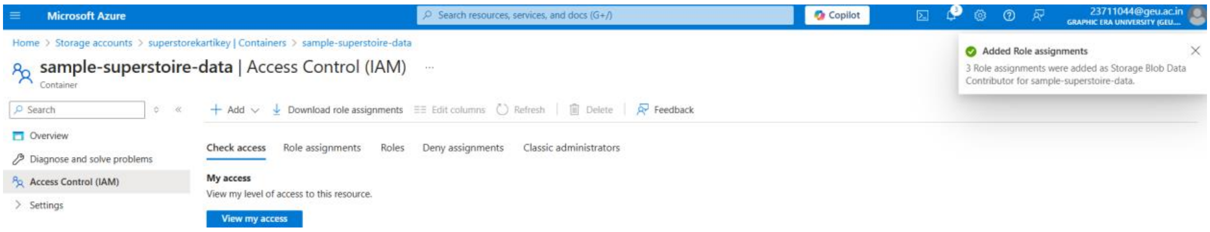


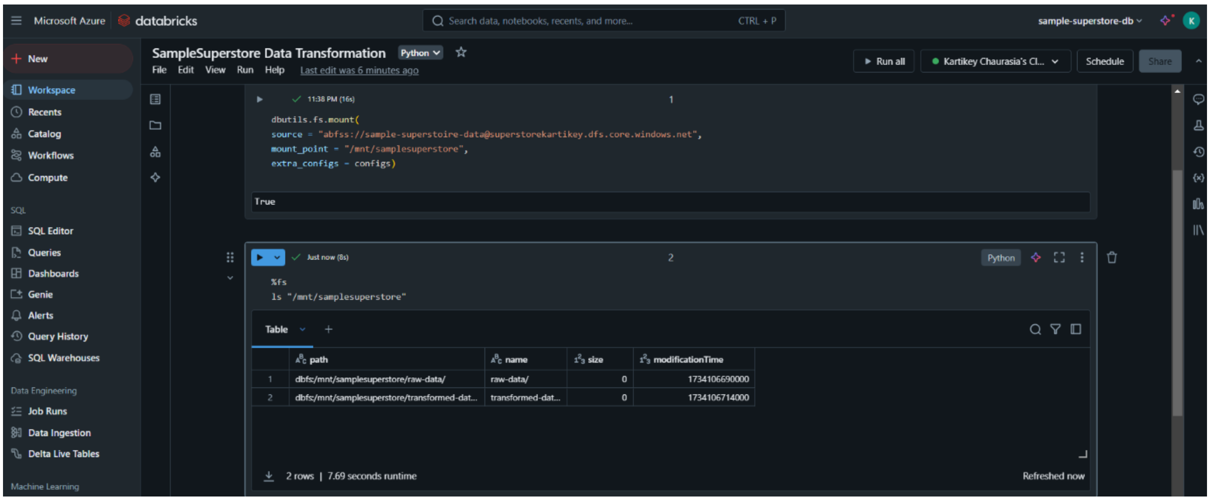



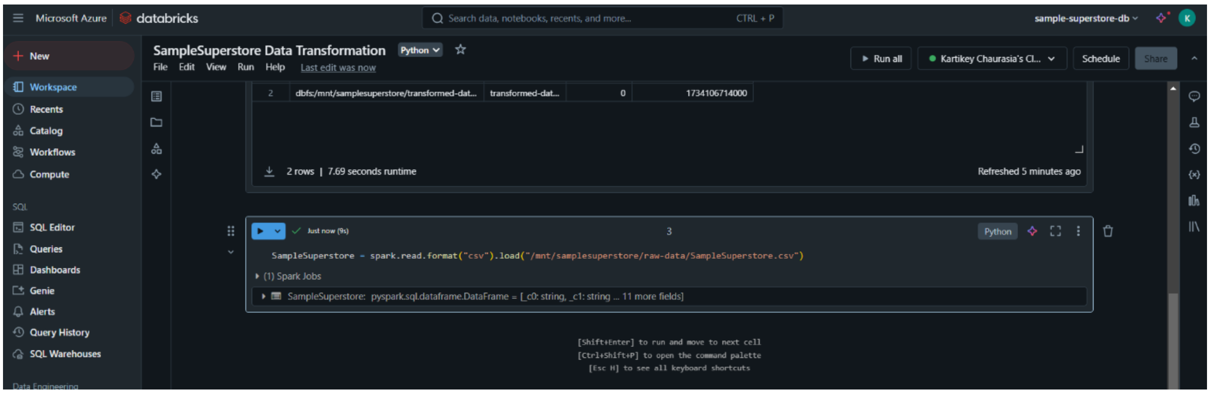


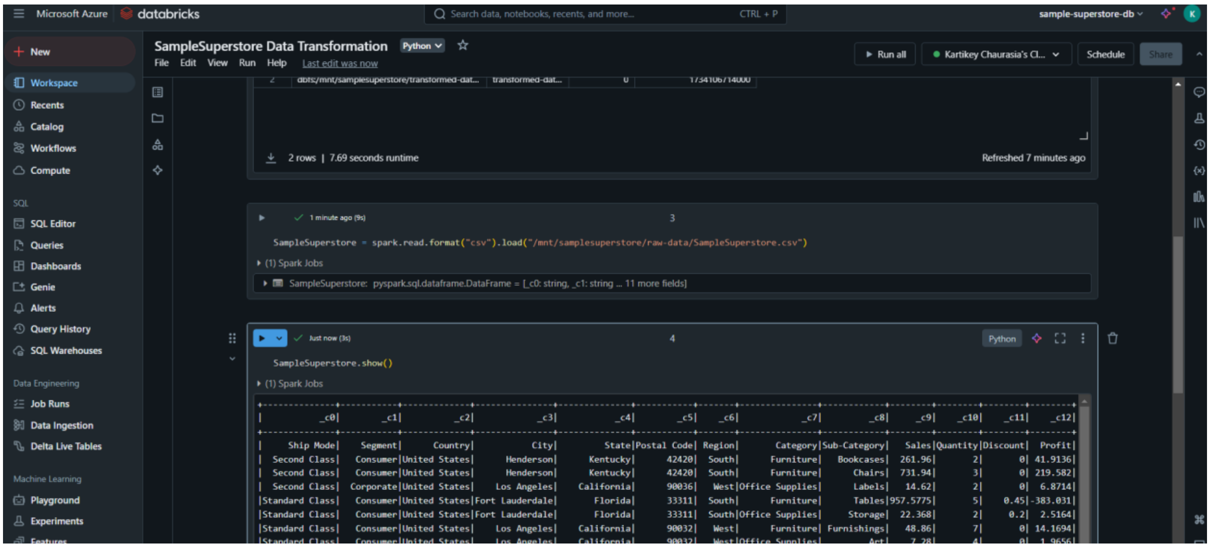



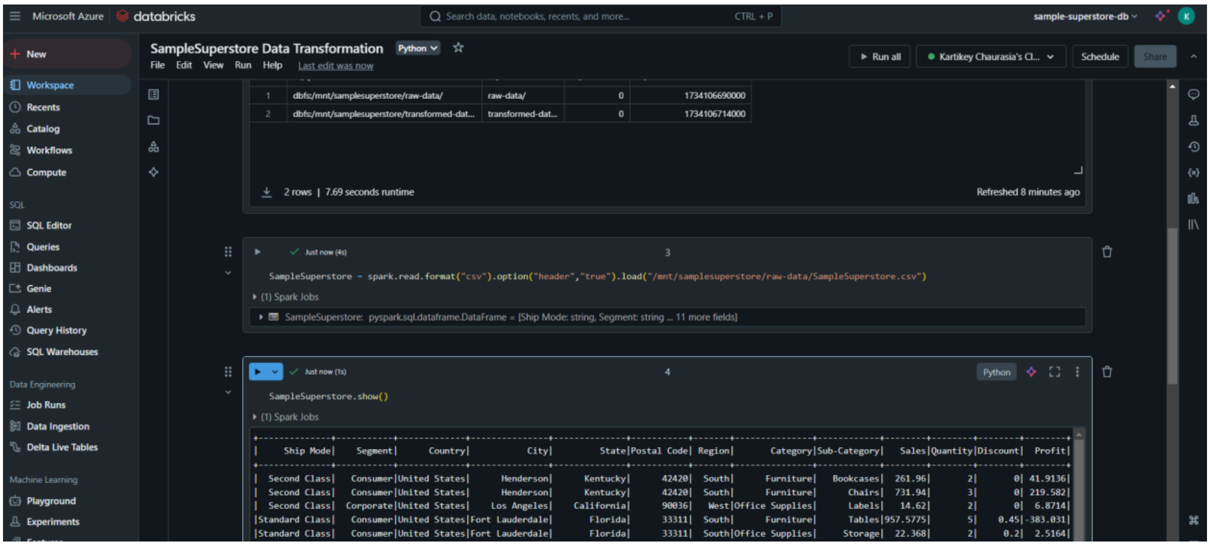

- Use Spark DataFrame APIs for reading, transforming, and writing data.





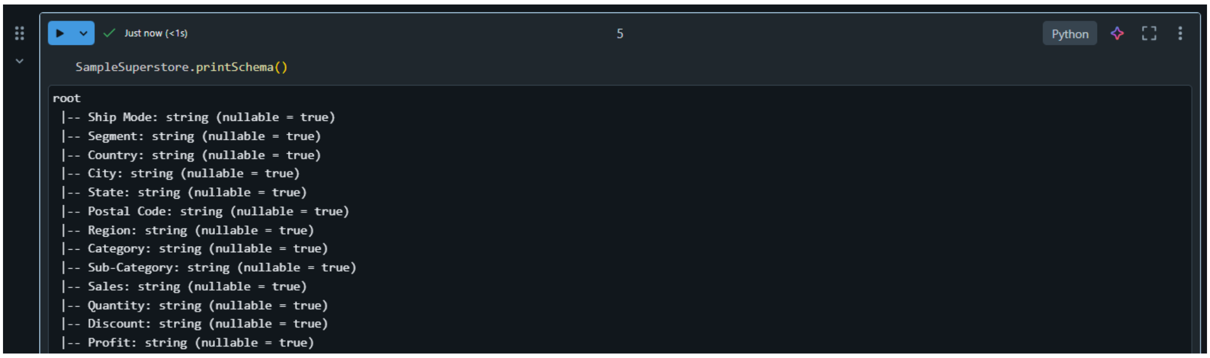

3. **Transformation Activities:**
   - Handle missing data by filling or removing null values.
   - Remove duplicate rows to ensure data integrity.
   - Calculate derived metrics, such as Profit Margin using formula-based transformations.

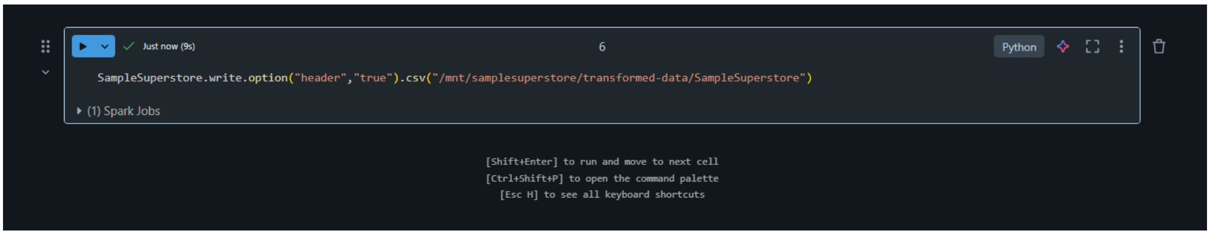


4. **Write Processed Data:**
   - Save the transformed dataset back to Data Lake Gen 2 or directly into Synapse Analytics for further analysis.

## Step 4: Data Analysis Using Azure Synapse Analytics

1. **Setup Synapse SQL Pools:**
   - Configure Synapse Analytics SQL pools to optimize data storage and retrieval.
   - Load the transformed dataset into Synapse tables.
2. **SQL Queries:**
   - Perform detailed analysis using SQL queries to identify sales trends, profitability, and geographic performance.



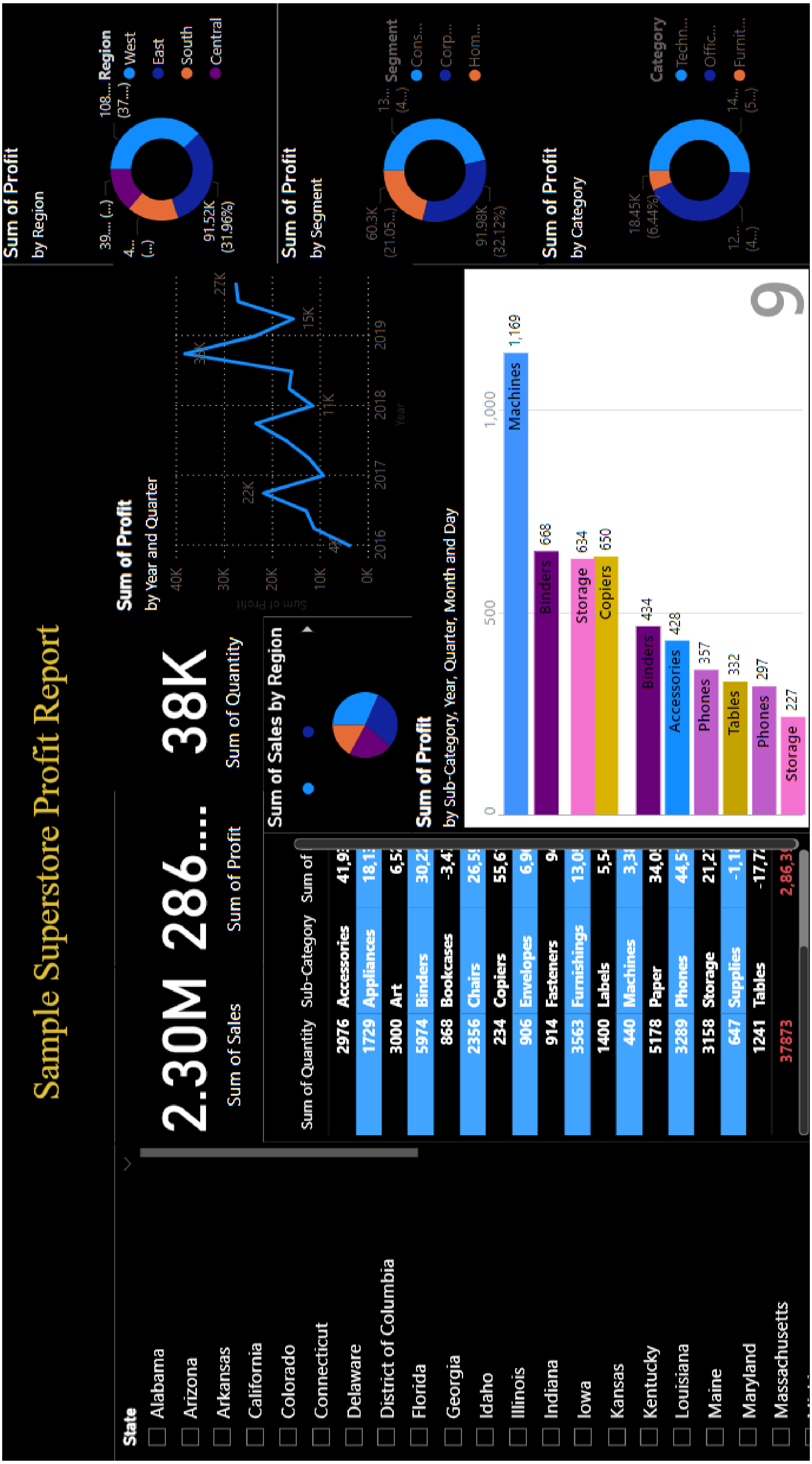




3. **Performance Optimization:**
   - Create indexes and materialized views to enhance query execution speed.

## Step 5: Visualization Using Power BI

1. **Connect Power BI to Synapse Analytics:**
   - Use the native Azure Synapse connector in Power BI Desktop to access processed data.
2. **Dashboard Creation:**
   - Design dashboards featuring:
     - Total and regional sales metrics.
     - Profitability trends across product categories.
     - Comparative analysis by customer segments and regions.
3. **Sharing Insights:**
   - Publish the dashboard to the Power BI Service.
   - Configure access controls and sharing permissions for stakeholders.

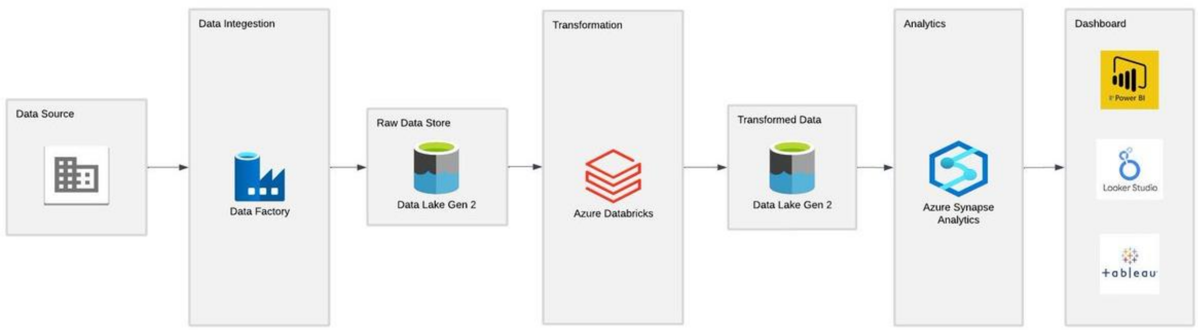
# ***Power BI Dashboard***


Sample Superstore Profit Report
2.30M
Sum of Sales
286....
Sum of Profit
38K
Sum of Quantity
State
Alabama
Arizona
Arkansas
California
Colorado
Connecticut
Delaware
District of Columbia
Florida
Georgia
Idaho
Illinois
Indiana
Iowa
Kansas
Kentucky
Louisiana
Maine
Maryland
Massachusetts
Sum of Profit
by Year and Quarter
Sum of Sales by Region
Sum of Profit
by Sub-Category, Year, Quarter, Month and Day
Sum of Profit
by Region
Region
West
East
South
Central
Sum of Profit
by Segment
Segment
Sum of Profit
by Category
Category
Machines
Binders
Storage
Copiers
Accessories
Phones
Tables

**Project Architecture Diagram**

The architecture of the project includes the following components:

- **Data Lake Gen 2:** Serves as the centralized storage for raw and transformed datasets.
- **Azure Data Factory:** Orchestrates data transfer and ensures workflows are automated.
- **Azure Databricks:** Performs distributed processing and complex data transformations.
- **Synapse Analytics:** Stores processed data and provides analytical capabilities.
- **Power BI:** Visualizes data and provides actionable insights to stakeholders.




**Challenges Faced and Solutions**

1. **Large Dataset Handling:**
   - **Challenge:** Initial slow query performance in Synapse Analytics.
   - **Solution:** Implemented data partitioning and indexing strategies to enhance performance.
2. **Access Issues:**
   - **Challenge:** Difficulty in connecting Azure services due to insufficient permissions.
   - **Solution:** Configured role-based access control (RBAC) and granted

appropriate permissions.

3. **Data Transformation Performance:**
   - **Challenge:** Slow processing of large datasets in Databricks.
   - **Solution:** Utilized optimized Spark configurations and parallel processing.

## Conclusion

This project highlights the integration of Azure cloud technologies to create an end-to-end data engineering solution. It ensures efficient data processing, robust analysis, and insightful visualization, offering scalable and reliable tools for modern business intelligence needs.

## Appendix

### Prerequisites

- Active Azure Subscription.
- Power BI Desktop installed on a local machine.
- Familiarity with SQL, Python, and Azure cloud services.